ROOT CAUSE ANALYSIS OF AUTOMOTIVE STRUCTURAL COMPONENT FAILURE

Case Study: Ford Front Subframe Recall (NHTSA 22V454)

A Materials & Quality Engineering Study

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ABSTRACT

This report presents a Root Cause Analysis (RCA) of the Ford front subframe bolt strip-out failure that led to NHTSA Safety Recall 22V454 in 2022. The failure involved a reduction in the yield strength of HSLA steel engine rail sub-assemblies from the specified minimum of 260 MPa to approximately 58 MPa, rendering the components incapable of withstanding assembly torque loads. The investigation revealed that a sub-tier supplier used an unapproved thermal e-coat stripping process, inadvertently causing severe annealing of the HSLA steel. This significantly reduced the material's dislocation density, altered its microstructure, a This report applies the Ishikawa Cause-and-Effect Diagram and 5-Why Analysis to identify the root cause and proposes corrective and preventive actions using the automotive industry's standard 8D problem-solving framework. The analysis highlights critical deficiencies in sub-tier supplier process change management, supplier oversight, and incoming inspection protocols. Recommendations are aligned with IATF 16949 automotive quality management requirements.

This study was undertaken as an independent academic project to develop practical skills in automotive quality engineering, materials failure analysis, and structured root cause investigation.

TABLE OF CONTENTS

1. INTRODUCTION

The automotive industry operates under some of the most stringent quality and safety standards of any manufacturing sector. Structural components such as subframes and chassis assemblies are subjected to dynamic loads throughout a vehicle's service life, making their mechanical integrity critical to occupant safety. Any deviation in material properties or manufacturing processes can have serious consequences, including safety recalls, financial liability, and reputational damage to the manufacturer.

This report investigates Ford Safety Recall 22V454, issued in 2022, which involved the front subframe assembly of vehicles produced at Ford's Chicago Assembly Plant. The recall was triggered by a bolt strip-out condition during assembly, which was subsequently traced to a significant reduction in the yield strength of HSLA steel engine rail sub-assemblies. The yield strength of affected components was found to be approximately 58 MPa, far below the design requirement of 260 MPa.

The objective of this report is to conduct a structured Root Cause Analysis of this failure using industry-standard quality tools including the Ishikawa Fishbone Diagram, 5-Why Analysis, and 8D Corrective Action framework. Additionally, this report provides a metallurgical explanation of the failure mechanism and proposes systemic preventive actions to avoid recurrence.

2. BACKGROUND

2.1 HSLA Steel in Automotive Applications

High Strength Low Alloy (HSLA) steels are widely used in automotive structural components due to their superior strength-to-weight ratio compared to conventional mild steels. These steels derive their mechanical properties from a combination of fine grain structure, solid solution strengthening through alloying elements such as niobium, vanadium, and titanium, and controlled thermomechanical processing during manufacturing. Typical yield strengths of automotive HSLA steels range from 260 MPa to 550 MPa depending on the grade. In structural applications such as subframes, engine rails, and chassis members, maintaining these mechanical properties is critical to vehicle safety and crashworthiness.

2.2 Electrocoating (E-coat) in Automotive Manufacturing

Electrocoating, commonly referred to as e-coat or electrodeposition coating, is a widely used corrosion protection process in the automotive industry. In this process, automotive components are submerged in a water-based paint bath and an electric current is applied, causing paint particles to deposit uniformly across the entire surface including recessed areas. E-coat provides excellent corrosion resistance and serves as the base layer before topcoat application. Occasionally, components that fail quality inspection must have their e-coat removed and reprocessed. The approved method for e-coat stripping is chemical stripping, which removes the coating without exposing the substrate metal to high temperatures.

2.3 Annealing and Its Effect on Steel Properties

Annealing is a heat treatment process in which a metal is heated to a temperature above its recrystallization point and subsequently cooled slowly. While annealing is intentionally applied in manufacturing to improve machinability and relieve residual stresses, accidental annealing of structural steel components has severe consequences. When HSLA steel is heated above its recrystallization temperature (approximately 550–700°C for low carbon steels), the existing dislocation network — which is responsible for the material's strength — is eliminated. Grains recrystallize and grow, resulting in a coarser, softer microstructure. The result is a dramatic reduction in yield strength, hardness, and fatigue resistance, rendering the component unfit for its intended structural application. In the case of Ford Recall 22V454, this accidental annealing reduced yield strength from 260 MPa to 58 MPa — a degradation of approximately 78%.

3. TIMELINE OF EVENTS

The following table presents the chronological sequence of events leading to the identification of the defect and subsequent issuance of Ford Safety Recall 22V454.

Date	Event
April 20, 2022	Ford Chicago Assembly Plant reports intermittent bolt strip-out condition during front subframe installation
April 2022	Tier-1 supplier Autokiniton Global Group (AGG) and Ford Supplier Technical Assistance (STA) notified to investigate root cause
April–May 2022	Sub-tier supplier analysis conducted on engine rail sub-assemblies; yield strength measured at 58 MPa against required ≥260 MPa
May 2022	Root cause confirmed — 656 engine rail sub-assemblies identified as having undergone unapproved thermal e-coat stripping, causing accidental annealing
May–June 2022	Ford initiates recall planning and scope determination for affected vehicles
2022	Ford Safety Recall 22V454 officially issued to NHTSA; free inspection and replacement offered to affected vehicle owners

The rapid escalation from assembly line detection to recall issuance reflects the safety-critical nature of structural component failures in automotive manufacturing.

4. ROOT CAUSE ANALYSIS

Root Cause Analysis (RCA) is a structured problem-solving methodology used in the automotive industry to identify the fundamental cause of a defect or failure, rather than addressing surface-level symptoms. In this investigation, two complementary tools are applied: the Ishikawa Cause-and-Effect Diagram to map all potential contributing factors, and the 5-Why Analysis to drill down to the single systemic root cause.

4.1 Ishikawa Cause-and-Effect Diagram

The following Ishikawa diagram maps all potential contributing factors to the subframe bolt strip-out failure across the six standard categories of the 6M framework.

Ishikawa Cause-and-Effect Diagram — Ford Recall 22V454

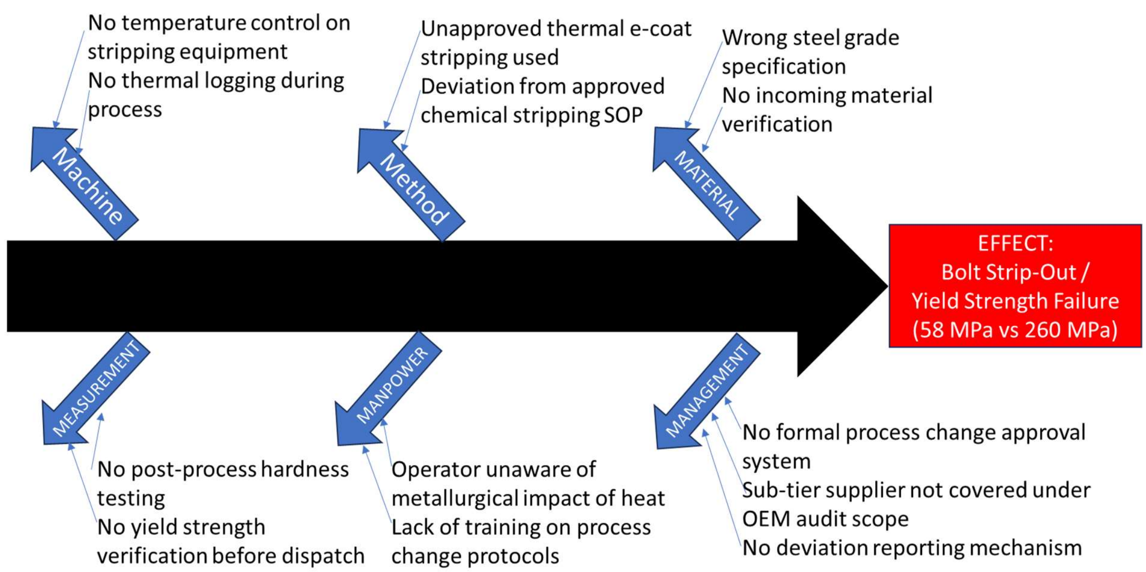

Figure 1: Ishikawa Cause-and-Effect Diagram for Ford Recall 22V454 Front Subframe Failure

4.2 Five-Why (5-Why) Analysis

The 5-Why technique was applied to systematically trace the failure from its observable symptom to its systemic root cause, as presented in the following table.

Why No.	Question	Answer
Why 1	Why did the bolt strip out during assembly?	The subframe material had insufficient yield strength (58 MPa vs required 260 MPa)
Why 2	Why was the yield strength so low?	The steel was accidentally annealed during processing, destroying its microstructure
Why 3	Why was the steel annealed?	An unapproved thermal e-coat stripping process was used instead of the approved chemical method
Why 4	Why was an unapproved process used?	There was no formal process change approval or notification system at the sub-tier supplier
Why 5	Why was there no process change control?	Ford's supplier audit programme did not extend coverage to sub-tier (Tier-2) manufacturing processes
Root Cause	—	**Absence of sub-tier supplier process change control and audit coverage by the OEM**

The 5-Why analysis confirms that while the immediate cause was a process deviation at the sub-tier supplier, the systemic root cause lies in the absence of formal process change governance across the supply chain.

5. MATERIALS ENGINEERING ANALYSIS

This section provides a metallurgical explanation of the failure mechanism observed in Ford Recall 22V454. The analysis examines the material properties of HSLA steel, the microstructural changes induced by accidental annealing, and the resulting mechanical property degradation that led to structural failure.

5.1 Material Identification

The engine rail sub-assemblies involved in this recall were fabricated from High Strength Low Alloy (HSLA) steel, a class of steel widely specified for automotive structural applications. HSLA steels typically contain small quantities of alloying elements including niobium (Nb), vanadium (V), and titanium (Ti), which promote grain refinement and precipitation hardening. These mechanisms collectively contribute to yield strengths in the range of 260–550 MPa depending on the specific grade. The design requirement of ≥260 MPa for the Ford engine rail sub-assemblies falls within the lower bound of standard automotive HSLA specifications, indicating that the material was selected with structural efficiency and weight optimisation in mind.

5.2 Microstructural Effect of Accidental Annealing

When HSLA steel is subjected to temperatures exceeding its recrystallization threshold — approximately 550–700°C for low carbon steels — significant and irreversible microstructural changes occur. The recrystallization process involves the nucleation and growth of new, strain-free grains at the expense of the existing deformed grain structure. In cold-worked or thermomechanically processed HSLA steels, the high dislocation density within the grain structure is the primary source of strengthening. Upon heating above the recrystallization temperature, this dislocation network is eliminated, grains grow to a coarser equiaxed morphology, and the precipitation strengthening effect of microalloying elements is partially reversed. The net result is a material that is significantly softer, more ductile, and structurally weaker than its original condition. In this case, the unapproved thermal e-coat stripping process exposed the steel components to elevated temperatures without controlled thermal cycling, effectively performing a full anneal on safety-critical structural parts.

5.3 Mechanical Property Degradation and Failure Mechanism

The consequence of accidental annealing was a reduction in yield strength from the required ≥260 MPa to approximately 58 MPa — a degradation of approximately 78%. At this reduced strength level, the material was unable to sustain the torque loads applied during bolt tightening of the subframe assembly. When bolt torque is applied, the compressive and shear stresses generated at the thread interface exceeded the now-weakened material's yield strength, causing plastic deformation and thread strip-out. Under normal operating conditions, the same torque would have remained well within the elastic range of the original HSLA steel. This case exemplifies the critical relationship between manufacturing process control and final component mechanical integrity — a deviation of minutes in an unapproved thermal process translated directly into a structural safety failure affecting thousands of vehicles.

6. CORRECTIVE ACTIONS (8D FRAMEWORK)

The 8D (Eight Disciplines) problem-solving framework is the standard corrective action methodology adopted across the global automotive industry. It provides a structured approach to problem containment, root cause elimination, and recurrence prevention. The following table presents the 8D response to the Ford Recall 22V454 subframe failure.

Discipline	Title	Action
D1	Team Formation	Cross-functional team formed comprising Ford Supplier Technical Assistance (STA), Tier-1 supplier Autokiniton, and sub-tier supplier quality and engineering representatives
D2	Problem Description	Bolt strip-out during front subframe assembly; yield strength of engine rail sub-assemblies measured at 58 MPa against design requirement of ≥260 MPa; 656 affected assemblies identified
D3	Interim Containment	100% quarantine and hardness testing of all suspect sub-assemblies; affected vehicles at assembly plant held pending inspection; immediate suspension of thermal stripping process at sub-tier supplier
D4	Root Cause Identification	Unapproved thermal e-coat stripping procedure annealed HSLA steel, eliminating dislocation strengthening and reducing yield strength by 78%; absence of process change approval system enabled deviation
D5	Corrective Actions	Mandatory post-stripping hardness and yield strength verification introduced; thermal stripping permanently prohibited in process specification; chemical stripping reinstated as sole approved method
D6	Implementation	Process Control Plan (PCP) updated to include in-process hardness check; incoming inspection criteria revised to mandate mechanical property verification for structural components
D7	Preventive Actions	Ford supplier audit programme extended to cover Tier-2 manufacturing processes; mandatory Process Change Notification (PCN) system implemented across all sub-tiers; IATF 16949 compliance verified at sub-tier level
D8	Team Recognition	Lessons learned documented and circulated across all Ford structural component suppliers; case study incorporated into supplier quality training programme

The 8D corrective actions address both the immediate containment of affected components and the systemic supply chain governance gaps identified as the root cause of this failure.

7. CONCLUSION

This report has presented a comprehensive Root Cause Analysis of the Ford front subframe bolt strip-out failure that resulted in NHTSA Safety Recall 22V454 in 2022. Through the application of Ishikawa cause-and-effect analysis and 5-Why methodology, the root cause was identified as the absence of sub-tier supplier process change control, which permitted an unapproved thermal e-coat stripping procedure to be performed on safety-critical HSLA steel structural components.

From a materials engineering perspective, the thermal stripping process inadvertently annealed the steel, eliminating the dislocation strengthening mechanisms responsible for its mechanical properties. The resulting yield strength degradation from 260 MPa to 58 MPa — a reduction of approximately 78% — rendered the components structurally inadequate for their intended application, directly causing the assembly line failure.

This case illustrates that automotive quality failures are rarely attributable to a single cause. The defect originated at the Tier-2 supplier through an unapproved process deviation, propagated through the Tier-1 supplier's incoming inspection gap, and was enabled by the OEM's limited sub-tier audit coverage. This multi-tier failure chain underscores the importance of robust supply chain quality governance, as mandated by the IATF 16949 automotive quality management standard.

The 8D corrective actions proposed in this report address both immediate containment and long-term systemic prevention. Key recommendations include mandatory mechanical property verification at incoming inspection, prohibition of thermal stripping in process specifications, and extension of OEM audit coverage to Tier-2 suppliers with mandatory Process Change Notification protocols.

This study was conducted as an independent academic exercise to develop practical competency in automotive quality engineering tools including RCA, FMEA-adjacent analysis, and 8D problem solving — skills directly applicable to quality and materials engineering roles in the automotive sector.

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